

PAPER_S204 — Session 204 Backfill:
Phase H204 Gap-Closure Footprint
(GW Inspiral, CP2 Class Registration, QCalc Ug-Sum)

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May 17, 2026

Abstract

Session 204 was a gap-closure pass. Three concrete defects in the UQFF calculator suite were repaired: (i) eleven new calculator classes were registered in `CondensedPhysics2.py`; (ii) the `QCalc.UnifiedFieldSolver` four-term *Ug-sum* was restored from an identically-zero stub; (iii) the `BlackHolePairsCalculator` placeholder value `term1 = 3.49e-59` was eliminated and replaced with dynamic gravitational-wave inspiral parameters from the Peters–Mathews formalism (chirp mass, quadrupole power, time-to-merger). This paper records the six exact-arithmetic identities (the H204 closures) that pin those repairs in the closure ledger. Three of the six are closed-form algebraic theorems (H204-1, H204-2, H204-6); two are definitional arithmetic checkpoints (H204-3, H204-5); one is an explicitly-labelled boolean post-condition (H204-4). The Yang–Mills mass gap is *not* claimed.

1 Ledger Summary

ID	Closure	Value	Class
H204-1	GW chirp mass, equal-mass binary, M_{chirp}/m	$2^{-1/5} \approx 0.8706$	THEOREM
H204-2	GW quadrupole coefficient, equal-mass binary	$64/5 = 12.8$	THEOREM
H204-3	CP2 gap-closure operation count	$11 \times 3 = 33$	arithmetic
H204-4	BlackHolePairs placeholder eliminated (boolean)	1	CONVENTION
H204-5	Ug-sum coupling-space dimension	4	arithmetic
H204-6	GW time-to-merger ratio for $m_A = 2m_B$	$1/8 = 0.125$	THEOREM

All six register as EXACT in `master_closures.csv` / `sigma_table.csv` (script: `_session204_closures.py`; final audit line: S204-GW-chirp-equal-mass: 0.870550563296 vs 0.870550563296 -> EXACT). Tier 11 of `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py` carries the six `record(...)` entries with full provenance.

2 Gravitational-Wave Inspiral Theorems (H204-1, H204-2, H204-6)

The CP2 `BlackHolePairsCalculator` now emits Peters–Mathews inspiral observables. Three exact identities follow from that formalism for equal-mass binaries and are recorded here as theorems.

2.1 Theorem H204-1 — Chirp Mass for Equal-Mass Binary

Theorem 1. *Let $m_1, m_2 > 0$ be the component masses of a compact binary. The Peters–Mathews chirp mass is*

$$M_{\text{chirp}} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}.$$

For $m_1 = m_2 = m$, $M_{\text{chirp}}/m = 2^{-1/5}$.

Proof. Substituting $m_1 = m_2 = m$:

$$M_{\text{chirp}} = \frac{(m \cdot m)^{3/5}}{(2m)^{1/5}} = \frac{m^{6/5}}{2^{1/5} m^{1/5}} = \frac{m^{6/5-1/5}}{2^{1/5}} = m \cdot 2^{-1/5}.$$

Hence $M_{\text{chirp}}/m = 2^{-1/5} \approx 0.8705505632961241$. The identity is algebraic in m and independent of G, c . $\square$

2.2 Theorem H204-2 — Quadrupole Coefficient for Equal-Mass Binary

Theorem 2. *The Peters (1964) quadrupole radiation power for a circular binary with component masses m_1, m_2 at orbital separation r is*

$$P_{\text{GW}} = \frac{32}{5} \frac{G^4}{c^5} \frac{m_1^2 m_2^2 (m_1 + m_2)}{r^5}.$$

For $m_1 = m_2 = m$,

$$\frac{P_{\text{GW}}}{(G^4 m^5)/(c^5 r^5)} = \frac{64}{5} = 12.8.$$

Proof. With $m_1 = m_2 = m$, the mass factor reduces to $m_1^2 m_2^2 (m_1 + m_2) = m^2 \cdot m^2 \cdot 2m = 2m^5$. Hence

$$P_{\text{GW}} = \frac{32}{5} \cdot \frac{G^4}{c^5} \cdot \frac{2m^5}{r^5} = \frac{64}{5} \cdot \frac{G^4 m^5}{c^5 r^5}.$$

Dividing both sides by $(G^4 m^5)/(c^5 r^5)$ gives the dimensionless coefficient 64/5. $\square$

2.3 Theorem H204-6 — Time-to-Merger Mass-Ratio Identity

Theorem 3. *The Peters (1964) time-to-merger for a circular binary with equal masses $m_1 = m_2 = m$ at fixed orbital separation r is*

$$t_{\text{merge}}(m, r) = \frac{5}{256} \frac{c^5 r^4}{G^3 m_1 m_2 (m_1 + m_2)} = \frac{5}{256} \frac{c^5 r^4}{G^3 2m^3}.$$

Comparing two such systems with equal-mass parameters m_A and m_B at the same r ,

$$\frac{t_{\text{merge}}(m_A, r)}{t_{\text{merge}}(m_B, r)} = \left(\frac{m_B}{m_A} \right)^3.$$

In particular, $m_A = 2m_B \Rightarrow t_A/t_B = 1/8 = 0.125$.

Proof. The constants $5/256$, c^5 , r^4 , G^3 , and the factor of 2 all cancel in the ratio, leaving $1/m_A^3 \div 1/m_B^3 = (m_B/m_A)^3$. For $m_A/m_B = 2$ the ratio is $(1/2)^3 = 1/8$. This identity is independent of G , c , and r . $\square$

3 Structural Checkpoints and Convention (H204-3, H204-4, H204-5)

3.1 H204-3 — CP2 Operation Count

Session 204 added eleven calculator classes to `CondensedPhysics2.py`: `VSDVPBSSHybridBlendCalculator`, `YangMillsDVPMassGapCalculator`, `BSFGWormholeTraversabilityCalculator`, `NuclearUmJWSTSynthesisCalculator`, `KozimaSCmCrossSectionCalculator`, `SCmActivationFunctionCalculator`, `BuoyancyLagrangianEOMCalculator`, `UQFFLagrangianDerivationCalculator`, `QCalcGeomVSDVPBSHCalculator`, `WolframExtractedPhysicsBridgeCalculator` and `VDSLENRIIsotopicEvolutionCalculator`. Each is verified by three operations (*import* / *instantiate* / *compute*); the total is $11 \times 3 = 33$. This is definitional arithmetic and a structural checkpoint matching `test_session204_gaps.py`; **it is not a theorem**.

3.2 H204-4 — Placeholder-Elimination Convention

Convention 4. The literal substring `'term1': 3.49e-59` is absent from `CondensedPhysics2.py`, and the key `'chirp_mass_kg'` is present. The closure value is 1 when both conditions hold.

This is a boolean edit footprint, not a derived mathematical identity. It is recorded so the gap-closure cannot regress silently and is explicitly labelled **CONVENTION** in the Tier-11 `record(...)` entry.

3.3 H204-5 — Ug-Sum Coupling-Space Dimensionality

Proposition 5. *Let*

$$g_{\text{Ug,sum}}(k_1, k_2, k_3, k_4) = \sum_{i=1}^4 k_i U_{g,i},$$

with $U_{g,1}, \dots, U_{g,4}$ functions of the system state independent of the couplings k_i . Then $g_{\text{Ug,sum}}$ is linear in each k_i and $\partial_{k_j} g_{\text{Ug,sum}} = U_{g,j}$, so $\sum_{j=1}^4 1 = 4$ counts the independent coupling dimensions.

The proposition is elementary linearity; it is registered because it is the precise dimensionality condition the corrected QCalc Ug-block now satisfies, replacing the previous identically-zero stub.

4 Honest Classification

- **Three closed-form algebraic theorems** (H204-1, H204-2, H204-6): proved here in full, independent of fundamental constants.
- **Two definitional arithmetic checkpoints** (H204-3, H204-5): counting identities, not theorems.
- **One explicitly-labelled convention** (H204-4): boolean post-condition of an edit.

What is *not* claimed. The `YangMillsDVPMassGapCalculator` added in this session computes calibrated quantities within the SCm parameter family. **This paper does not promote the Yang–Mills mass gap to a proven Clay–Millennium result.** The new calculator is a UQFF-internal predictor; the Clay problem remains open.

5 Reproducibility

1. Closure script: `_session204_closures.py`. Final parseable line:
`S204-GW-chirp-equal-mass: 0.870550563296 vs 0.870550563296 -> EXACT.`
2. Audit refresh: `python _uqff_program.py -audit -sigma`. Row 4 of `master_closures.csv` and `sigma_table.csv` now carries the `S204-GW-chirp-equal-mass EXACT` entry.
3. Tier-11 `record(...)` entries: six rows tagged `H204-*` in `unified_closure_audit.json`, all `status: DERIVED`, paper field `PAPER_S204` sec 2/3.
4. Independent verification: the chirp-mass, quadrupole-coefficient, and time-to-merger identities are re-derived symbolically with `sympy` using a code path orthogonal to the closure script.